

Adarsh Gadari

Machine Learning Research Engineer

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| Pittsburgh, PA

Professional Summary

Machine Learning Research Engineer with 9+ years of experience translating AI research into production clinical systems. Demonstrated ability to take work from conception through deployment — including three revenue-generating ophthalmic AI applications on the Heidelberg Engineering ecosystem, used across clinical trials and active patient care networks, and a label-free retinal segmentation system independently built and deployed on UPMC MyApps. Research spans computer vision, multimodal deep learning, and molecular generative modeling, with active work on large-scale polymer generation using transformer architectures trained on 100M-compound datasets. Brings a rare combination of rigorous research credibility — including peer-reviewed publications, 25+ ARVO poster contributions, and session moderation at the Retina in AI track — with the engineering discipline to build FDA 21 CFR Part 11-compliant systems that reach clinicians.

Experience

Research Collaborator — Molecular Generative Modeling

Jan 2026 – Present

Old Dominion University

Remote

- Collaborating on a large-scale polymer generative modeling project using a GPT-based transformer trained on SMILES representations from a 100M-polymer dataset (PolyOne), with training executed on Google Cloud Vertex AI using distributed GPU infrastructure.
- Preliminary runs on 1M and 10M-polymer subsets demonstrate encouraging validity, novelty, and uniqueness metrics across both conditional and unconditional generation objectives; actively scaling toward the full dataset. Manuscript in preparation.

Lead Researcher

2025 – Present

Netramind Innovations

Pittsburgh, PA

- Leading and mentoring graduate students, undergraduate researchers, and newly onboarded engineers at Netramind toward efficient use of deep learning models under label-free and pseudo-label frameworks.
- Developed deep learning models within a pseudo-labeling framework for robust, accurate segmentation of retinal layers in complex OCT images.
- Developed an automated method using hierarchical binarization and deep learning to calculate the Choroidal Vasculature Index (CVI) by extracting complex vascular structures.
- Deployed AI-driven CVI software on the Heyex platform — the ophthalmic imaging OEM software ecosystem by Heidelberg Engineering GmbH.
- Deployed 3 production-grade, public-facing clinical AI applications within one year on the Heidelberg Engineering (HE) AppWay ecosystem — owning the full stack from model training and backend development to Azure deployment, HE integration, and iterative collaboration with the HE engineering team to resolve technical and compliance concerns through to launch.
- **CVI (Choroidal Vasculature Index):** Revenue-generating application for automated choroidal vascular analysis, deployed across the Heidelberg device network; processed 1,000+ OCT volumes across internal and external datasets. **NMI-ORA (Retinal Layer Segmentation):** Revenue-generating platform central to 3 completed clinical trials (2 Phase I, 1 Phase II) and 1 post-hoc analysis spanning 680+ OCT volumes — contributing not only to algorithm development and deployment but to trial execution itself, collaborating directly with clinicians and graders to generate protocol-defined clinical endpoints per trial agreements; 1 ongoing multi-center study (30+ OCT volumes) in progress. **NMI-AMD-SC (AMD Referral Decision Support):** Deployed across a network of 10+ optometrists in Pennsylvania, providing evidence-based guidance on when to escalate AMD patients to a retinal surgeon.
- Conceived and led a multi-team research initiative establishing that device-generated Q-scores (Heidelberg IQ metric) are not reliable indicators of deep learning segmentation performance — providing empirical proof across

5,000+ OCT B-scans that the correlation between IQ score and segmentation quality is negligible; currently proposing task-specific quality metrics as a replacement framework. Manuscript in preparation for submission to *Nature Digital Medicine* or *Computers in Biology and Medicine*.

- Driving adoption of a Quality Management System (QMS) aligned with GCP and FDA 21 CFR Part 11 standards, establishing version control workflows, project management infrastructure, and audit-ready documentation practices to support compliant software development across the team.

Researcher (Fellow, Senior and Professional)

University of Pittsburgh / UPMC

2022 – Present

Pittsburgh, PA

- Conceived, designed, and solely implemented a novel label-free deep learning framework to segment retinal layers on the widely used Cirrus OCT device — without any manual annotations — producing retinal layer thicknesses, ETDRS-grid thickness maps, and clinical color maps; the system is now deployed on UPMC MyApps and actively used by clinicians across the health network.
- The label-free training paradigm developed through this research directly enabled Netramind Innovations to build and commercialize NMI-ORA — a revenue-generating retinal segmentation product now used in clinical trials — establishing a new framework for annotation-free deep learning in ophthalmic imaging.
- Collaborated with the UPMC enterprise team to productionize the framework: refactored the inference pipeline using ONNX and workflow optimizations, reducing end-to-end analysis time from 12 minutes to 90 seconds on 128-scan volumes (8× speedup on GPU); participated in architecture and code reviews, contributed directly to production code, and served as the primary technical resource for onboarding the enterprise team to the codebase.
- Collaborating with a multidisciplinary clinical team — comprising medical students, residents, and retinal surgeons — to design and execute clinically relevant studies that leverage AI algorithms for causal relationship discovery in ophthalmic imaging data.
- Investigating the relationship between ellipsoid zone (EZ) thickness, broader retinal layer thickness profiles, and a newly developed contrast sensitivity metric, with the goal of establishing quantitative links between structural biomarkers and functional visual outcomes.
- Quantifying a recently FDA-approved biomarker for drug discovery and disease monitoring by analyzing complex ellipsoid zone structures from OCT volumetric imaging data.
- Researching out-of-distribution identification for AI models to characterize potential failure modes within the ophthalmic AI ecosystem.
- Contributed to 14 poster presentations at ARVO 2025 (Salt Lake City, UT) showcasing AI-based ophthalmic imaging tools developed for medical students' clinical studies.
- Presenting 11 posters and serving as Session Moderator for the *Retina in AI* poster session at ARVO 2025 (May 2–7, Denver, CO).

Graduate Assistant (Teaching and Research)

University of North Carolina at Greensboro

2022 – 2025

Greensboro, NC

- Investigated domain adaptation techniques and structurally informed deep learning models to enhance model robustness across varying image characteristics.
- Managed 3+ courses per semester (class sizes 10–40 students) covering Deep Learning, Image Processing, and Algorithm Analysis; served as Summer 2023 Instructor for Elementary Data Structures and Algorithm Analysis.

AI/ML R&D Software Engineer

Carelon Global Solutions (formerly Legato Health Technologies · Subsidiary of Elevance Health)

2020 – 2022

Hyderabad, India

- Developed a ticket bucketing classification system using text embeddings, enabling automation that reduced manual effort and achieved \$100K in annual cost savings.
- Proposed and designed multiple AI-driven features for tele-health platforms to enhance communication between clinicians and individuals with disabilities.
- Developed CNN-based algorithms fusing multimodal signals to assess patient readmission risk as a proof-of-concept for predictive healthcare analytics.
- Performed clustering and identification of high-risk patient groups from demographic data to enhance care delivery and optimize revenue outcomes.

Research Fellow

LV Prasad Eye Institute

2019 – 2020

Hyderabad, India

- Developed an intra-ocular lens (IOL) power prediction model using structured clinical data and machine learning (Random Forest Regressor), achieving performance comparable to industry-standard IOL formulas.
- Built a custom OCR pipeline to extract and statistically analyze data from a large cohort of patient PDF records.

Data Scientist

Cyient Insights

2018 – 2019

Hyderabad, India

- Developed proof-of-concept models for failure prediction and anomaly detection under class-imbalanced settings for international clients.
- Built Tableau dashboards for business intelligence and operational insights.

Education

Master of Science in Computer Science

University of North Carolina at Greensboro

Aug 2022 – May 2025

Greensboro, NC

- **Thesis:** Exploring Explainability of Hypergraph Neural Networks using Mixed-Integer Programming.

Master of Technology in Electrical Engineering

Indian Institute of Technology Hyderabad

May 2018

Hyderabad, India

- **Thesis:** Towards True 3D Display: Technological Feasibility of a Flatscreen Solution.

Bachelor of Engineering in Electronics and Communication

Osmania University

May 2015

Hyderabad, India

Technical Skills

- **AI/ML:** Deep Learning, Foundational Models, Generative AI, LLMs, VLMs, Representation Learning, Multimodal Networks, Anomaly Detection, Segmentation, Supervised/Unsupervised/Self-Supervised Learning, Hypergraph Neural Networks, Explainability
- **Signal & Image Processing:** Computer Vision, OCT Imaging, Frequency Domain Analysis, NLP
- **Programming:** Python (PyTorch, TensorFlow, Gurobi), C, C++, MATLAB
- **Cloud & Deployment:** Azure, AWS, Docker, Google Cloud (Vertex AI)
- **Tools:** Git, GitHub, Bitbucket, Jira, Confluence, Linux/Windows/macOS
- **Project Management:** Project Initiation, Work Breakdown Structure, RACI Charts, Agile/Scrum

Selected Publications

- Hasan, N. *, **Gadari, A. ***, et al. “Deep Learning–Based Automated Segmentation and Quantification of Ellipsoid Zone and RPE–Bruch’s Membrane Complex in Healthy Subjects and Geographic Atrophy.” *Preprints* (2026). [doi:10.20944/preprints202604.1517.v1](https://doi.org/10.20944/preprints202604.1517.v1) *Equal contribution
- **Gadari, Adarsh**, et al. “Robust Retinal Layer Segmentation Using OCT B-Scans: A Novel Approach Based on Pix2Pix Generative Adversarial Network.” *14th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics*, 2023.
- **Gadari, Adarsh**, et al. “Retinal Sublayer Segmentation Based on Deep Learning Using Optical Coherence Tomography B-Scans: Training Multilayer Masks vs Boundary Labels.” *Investigative Ophthalmology & Visual Science* 65.7 (2024): 2408–2408.
- Chhablani, Jay, **Gadari, Adarsh**, et al. “Deep-Learning-Based Choroid Vessel Segmentation Using Feature Maps of ResUNet GAN Autoencoder Architecture.” *Investigative Ophthalmology & Visual Science* 66.8 (2025): 587–587.
- **Gadari, Adarsh**, et al. “Synthetic Ground-truth Image-based Drusen Segmentation in OCT Images: A Novel Approach Based on Sinusoidal Image Deformations and Generative Adversarial Networks.” *Investigative*

Ophthalmology & Visual Science 66.8 (2025): 4188–4188.

- **Gadari, Adarsh**, et al. “The Disconnect Between Device Quality Score and Deep Learning based Choroid Segmentation in Retinal OCT.” *ARVO Annual Meeting*, Denver, CO, May 2026. [Poster Presentation]

Certifications

- **Foundations of Project Management** — Google, Coursera (Apr 2025)
- **Project Initiation: Starting a Successful Project** — Google, Coursera (Jun 2025)